

ing the following commands one by one in the command prompt:

```
1) avrdude -p atmega128 -P usb -vvv
-c avrispmkII -U lfuse:w:0x0ce:m
2) avrdude -p atmega128 -P usb -vvv
-c avrispmkII -U hfuse:w:0xd9:m
3) avrdude -p atmega128 -P usb -vvv
-c avrispmkII -U efuse:w:0xff:m
```

The command for loading the hex code(webchart.hex) into the microcontroller is as follows:

```
avrdude -p atmega128 -P usb -vvv -c
avrispmkII -U flash:w:webchart.hex:i
```

## Construction and testing

An actual-size, single-side PCB for the circuit is shown in Fig. 5 and its component layout in Fig. 6.

All the components are easily available in electronics components shops. After assembling the components on PCB, you can connect the LCD. Ensure that power supply connections are properly checked. Connect the board and the PC using Ethernet cable. Switch on the circuit. LED3 at PC0 blinks and at the same time the LEDs (LED4 and LED5) near the ENC28J60 blink continuously. Depending upon the stack, LED3 may stop blinking sometimes since it is in polling loop. If all these LEDs are blinking, it means the circuit is working fine. Now, you can see 192.168.0.80 being displayed on the LCD.

Next, configure the network connection in your PC from the control panel. Note that basic knowledge of computer networking is required for configuration. Now, open Web browser on your PC. We have tested it on Google Chrome but it will work on Mozilla Firefox also. Type the IP address as 192.168.0.80 on the address bar. If everything is fine, you will see a window on your computer screen as shown in Fig. 7. This is the HTML main page for monitoring the water level in the overhead tank having five sensors, viz, A, B, C, D and SPI. Place the sensors as shown in the circuit diagram, with SPI sensor connected to +Vcc and place it near the bottom of the tank. The water in the tank is shown by graphical green colour



Fig. 7: Program screenshot

inside the tank image. When water level reaches D level, you need to turn off the motor by clicking 'motoroff' button.

Some of the parameters given under status section on the main page are for debugging purpose only. The 'Pot0: 5' displays analogue voltage at port PA1, the button displays the water-level sensor input status at port F. The LEDs display the output status of port C.

For connecting the target board to the Internet, connection through modem is necessary so as to make the board accessible through the Web page from any part of the world. We have already set the IP address at 192.168.0.80 in the firmware. If you have other 192.168.0.xx node setup, this arrangement is not required. By typing ipconfig in the command prompt, you can verify your node network address. Normally, the default IP address used by Linksys broadband routers is 192.168.1.1, which will be the gateway for the board. You can change your board IP address suiting your existing network in the code, but for that you have to recompile the code and burn the hex file again.

The prototype was tested with the board connected directly to PC using Ethernet cable (straight type). To test it on the local network, you can connect your computer to the same network via Wi-Fi or Ethernet and type 192.168.0.80 in the browser.

## EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download on [source.efymag.com](http://source.efymag.com) website.

You will get default home page with EFY logo along with other details as shown in Fig. 7. To access the board from outside the network, you need to apply port forwarding setting in the router.

On top of the main page, you can see 'features' and 'architecture' options. This is just additional information related to the projects. If there is display problem on the LCD, you can vary VR1 to change the contrast until the character is clearly visible. If problem persists, press reset switch S1 momentarily and check the display again. For more troubleshooting, you can refer to the test points table.

## Further applications

Monitoring of parameters such as temperature, pressure, voltage and current can be done via Web page from any part of the world by connecting suitable sensors to the input port of the microcontroller. ●

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